# MODULE - IV

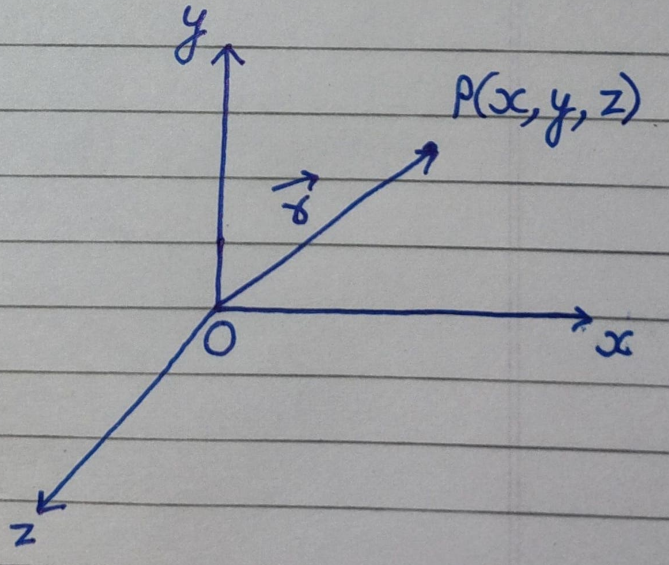
## Relativistic Mechanics

The classical mechanics can be applied if velocity of a particle is low ($v << c$), but it fails at $v \approx c$. Theory of Relativity can be applied for all velocities from 0 to c.

# Frame of Reference

A system of coordinate axes ~~which~~ with respect to which the position of a particle or event is described is called frame of reference. It must be rigid.

Example-1: Cartesian Coordinate System, in which three coordinates $x, y$ & $z$ are used to describe the position.




Position Vector,
$$\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$$

Velocity, $\vec{v} = \dfrac{d\vec{r}}{dt}$

$$\vec{v} = \hat{i}\frac{dx}{dt} + \hat{j}\frac{dy}{dt} + \hat{k}\frac{dz}{dt}$$

Acceleration, $\vec{a} = \dfrac{d\vec{v}}{dt} = \dfrac{d^2\vec{r}}{dt^2}$

$$\vec{a} = \hat{i}\frac{d^2x}{dt^2} + \hat{j}\frac{d^2y}{dt^2} + \hat{k}\frac{d^2z}{dt^2}$$

Example-2: Space Time Frame is a coordinate system with four coordinates $x, y, z$ & time '$t$' is called space-time frame.

Inertial Frame:

In this frame, Newton's laws are valid. This frame is at rest or moves with constant velocity.

Non-Inertial Frame:

In this frame, Newton's laws are not valid. They are either rotating or accelerating.

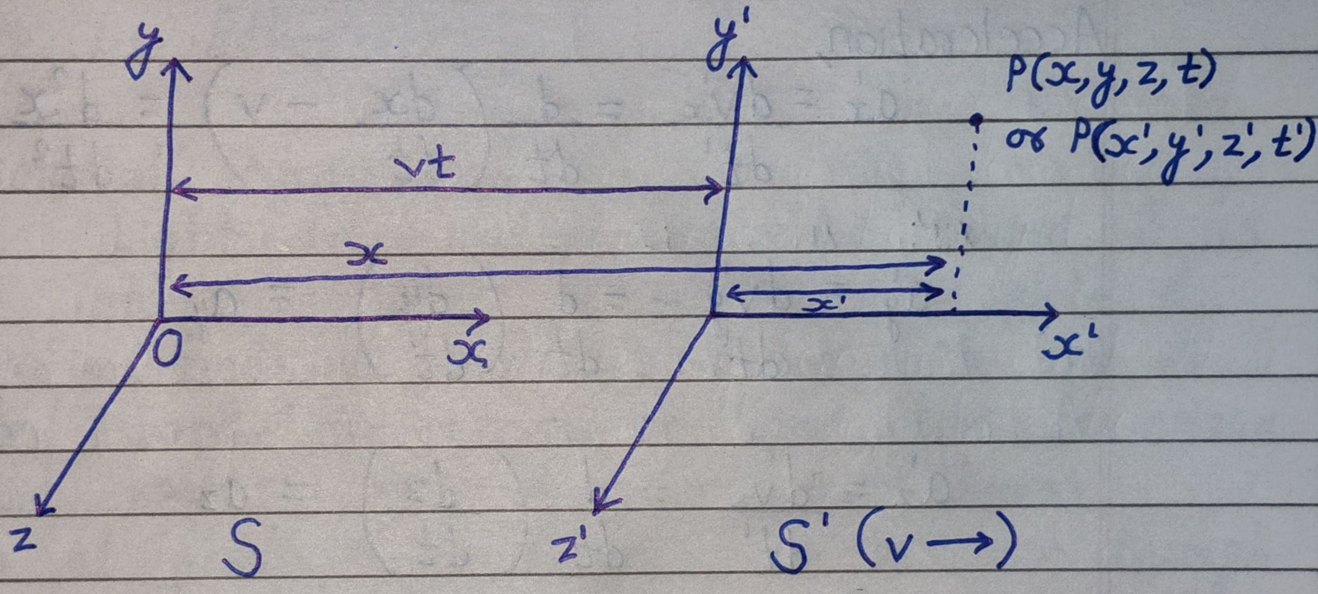
Q. Is Earth an Inertial Frame?

⇒ Earth is non-inertial frame because it rotates. Since, the speed is very low therefore it can be considered as inertial frame of reference.

# Galilean Transformation Equations

Consider two inertial frames S & S'. The equations relating two set of coordinates are called Tranformation Equation.

At $t = t' = 0$, both frames are coincident. As time passes, S is at rest and S' moves with constant velocity v in tve x-direction.




From diagram,

$$x = x' + vt$$

$$\Rightarrow x' = x - vt \quad \text{——— (1)}$$

$$y' = y \quad \text{——— (2)}$$

$$z' = z \quad \text{——— (3)}$$

$$t' = t \quad \text{——— (4)}$$

Velocity,

$$v_x' = \frac{dx'}{dt'} = \frac{d(x-vt)}{dt} = \frac{dx}{dt} - v$$

$$v_y' = \frac{dy'}{dt'} = \frac{dy}{dt}$$

$$v_z' = \frac{dz'}{dt'} = \frac{dz}{dt}$$

Acceleration,

$$a_x' = \frac{dv_x'}{dt'} = \frac{d}{dt}\left(\frac{dx}{dt} - v\right) = \frac{d^2x}{dt^2} - 0 = a_x$$

$$a_y' = \frac{dv_y'}{dt'} = \frac{d}{dt}\left(\frac{dy}{dt}\right) = a_y$$

$$a_z' = \frac{dv_z'}{dt'} = \frac{d}{dt}\left(\frac{dz}{dt}\right) = a_z$$

$$\Rightarrow \vec{a'} = \vec{a}$$

i.e. Acceleration ~~in~~ is invariant under Galilean Transformation.

# Michelson - Morley Experiment

Concept of Ether:

A hypothetical medium Ether was assumed to be present everywhere in space. Ether must be rigid, massless, invisible and transparent.

Imp. Objective:

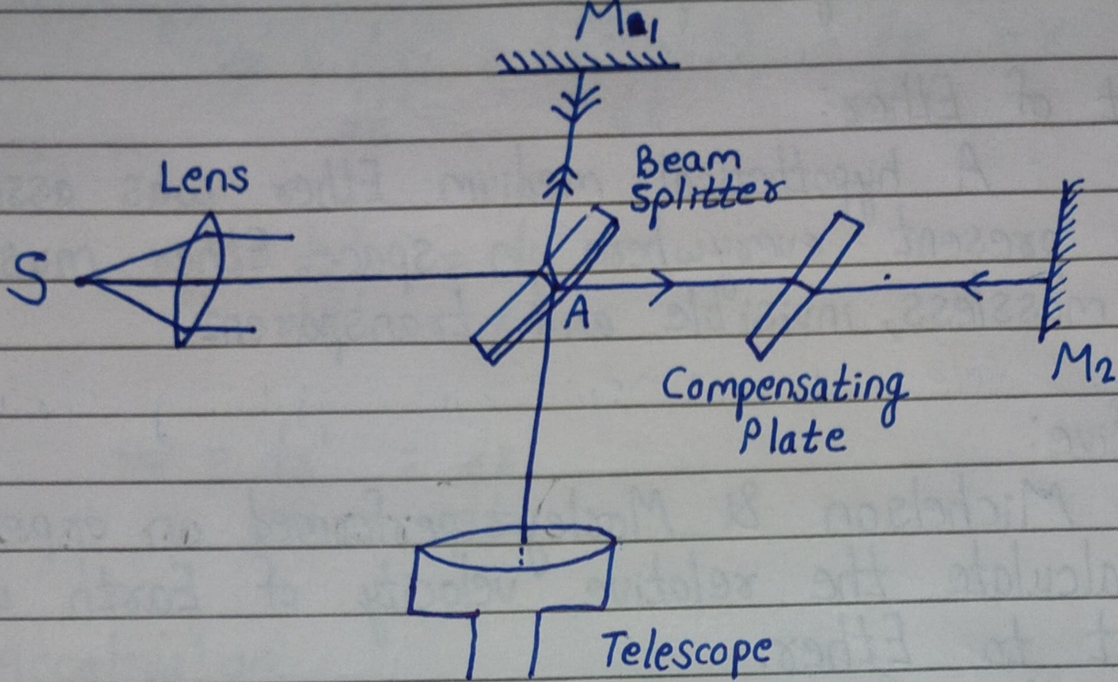
Michelson & Morley performed an experiment to calculate the relative velocity of Earth with respect to Ether.

Experimental Arrangement:

Light from mono-chromatic source S is divided into two parts by beam-splitter plate A. The reflected light travels to mirror $M_1$ and is reflected back. The transmitted light goes to mirror $M_2$ and reflects back. The two light beams overlap at point A so fringes can be seen through telescope.

The reflected light travels beam-splitter ~~twice~~ thrice while the transmitted light only ones. To compensate this path difference, a compensating plate is used which is identical to beam-splitter plate.

The whole assembly was mounted on a stone-slab floating in mercury.

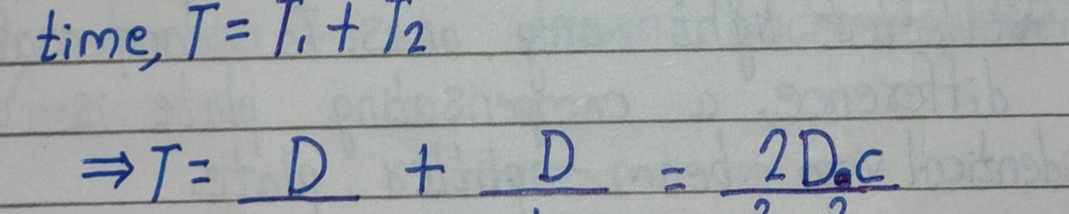



Theory:

From figure,

$$AM_1 = AM_2 = D$$

Let, $T_1$ is the time taken by light to travel from A to $M_1$ and $T_2$ to travel from $M_1$ to A.




Total time, $T = T_1 + T_2$

$$\Rightarrow T = \frac{D}{c-v} + \frac{D}{c+v} = \frac{2Dc}{c^2 - v^2}$$

Optical path, $x_1 = Tc$

$$\Rightarrow x_1 = \frac{2Dc^2}{c^2 - v^2}$$

$$\Rightarrow x_1 = \frac{2Dc^2}{c^2(1 - v^2/c^2)}$$

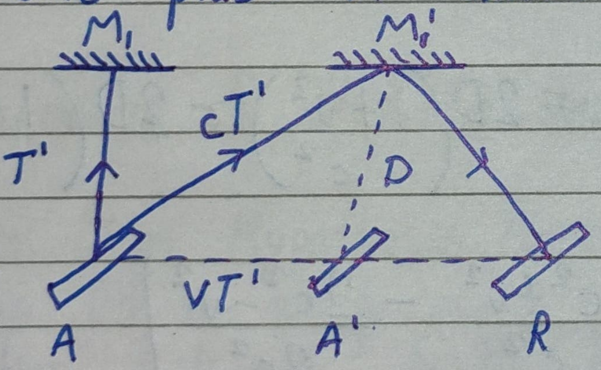
$$\Rightarrow x_1 = 2D\left(1 - \frac{v^2}{c^2}\right)^{-1}$$

Using Bionomial Theorem and Neglecting higher orders

$$\left[(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}.x^3 + \dots\right]$$

$$x_1 = 2D\left(1 + \frac{v^2}{c^2}\right) \quad \text{——— (1)}$$

Let, $T'$ be the time taken by light to travel from A to $M_1$. During this time, mirror moves to position $M_1'$ and plate A shifts to position $A'$.




$$c^2T'^2 = v^2T'^2 + D^2$$

$$\Rightarrow T'^2(c^2 - v^2) = D^2$$

$$\Rightarrow T'^2 = \frac{D^2}{c^2 - v^2}$$

$$\Rightarrow T' = \frac{D}{\sqrt{c^2 - v^2}}$$

Total time $= 2T'$

Optical path, $x_2 = 2T'c$

$$\Rightarrow x_2 = \frac{2Dc}{\sqrt{c^2 - v^2}}$$

$$\Rightarrow x_2 = \frac{2Dc}{c\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow x_2 = 2D\left(1 - \frac{v^2}{c^2}\right)^{-1/2}$$

Applying Binomial theorem and neglecting higher order terms,

$$x_2 = \left(1 + \frac{v^2}{2c^2}\right) . 2D \quad \text{——— (2)}$$

$$\text{Path diff.} = x_1 - x_2 = 2D\left(1 + \frac{v^2}{c^2}\right) - 2D\left(1 + \frac{v^2}{2c^2}\right)$$

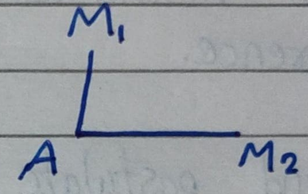
$$\Rightarrow \text{Path diff.} = 2D\left[\frac{c^2 + v^2}{c^2} - \frac{2c^2 - v^2}{2c^2}\right]$$

$$\Rightarrow \text{Path diff.} = \frac{2D}{2c^2}\left[2c^2 + 2v^2 - 2c^2 - v^2\right]$$

$$\Rightarrow \text{Path diff.} = \frac{Dv^2}{c^2}.$$

Now, the apparatus is rotated through 90°, so that optical path $AM_1$ is longer than $AM_2$ by the same amount. Therefore, total path diff. $= \frac{2Dv^2}{c^2}$.

If this path diff. corresponds to shifting of n-fringes, then $\frac{2Dv^2}{c^2} = n\lambda$.




For $\lambda = 5000$ Å, $D = 10$ m
$\Rightarrow n = 0.4$ fringes.

**Negative Result:**

Michelson & Morley expected a fringe shift of 0.4 fringes. But no fringe-shift was observed in any direction. The negative result of this experiment strike on the roots of Ether Hypothesis.

**Explanation of Negative Result:**

(i) **Ether Drag Hypothesis:** According to Michelson, the earth drags Ether along with it. There is no relative motion between Earth & Ether. Therefore, no fringe shift is observed.

(ii) **Lorentz-Fitz Gerald Contraction Hypothesis:** According to Lorentz, the length of a moving body is contracted by a factor $\sqrt{1-\frac{v^2}{c^2}}$ in the direction of motion.

Therefore, path diff. is zero and no fringe-shift is observed.

Imp. (iii) <u>Constancy of Speed of Light</u>: According to Einstein, speed of light is constant and does not depend upon motion of source, medium or observer.

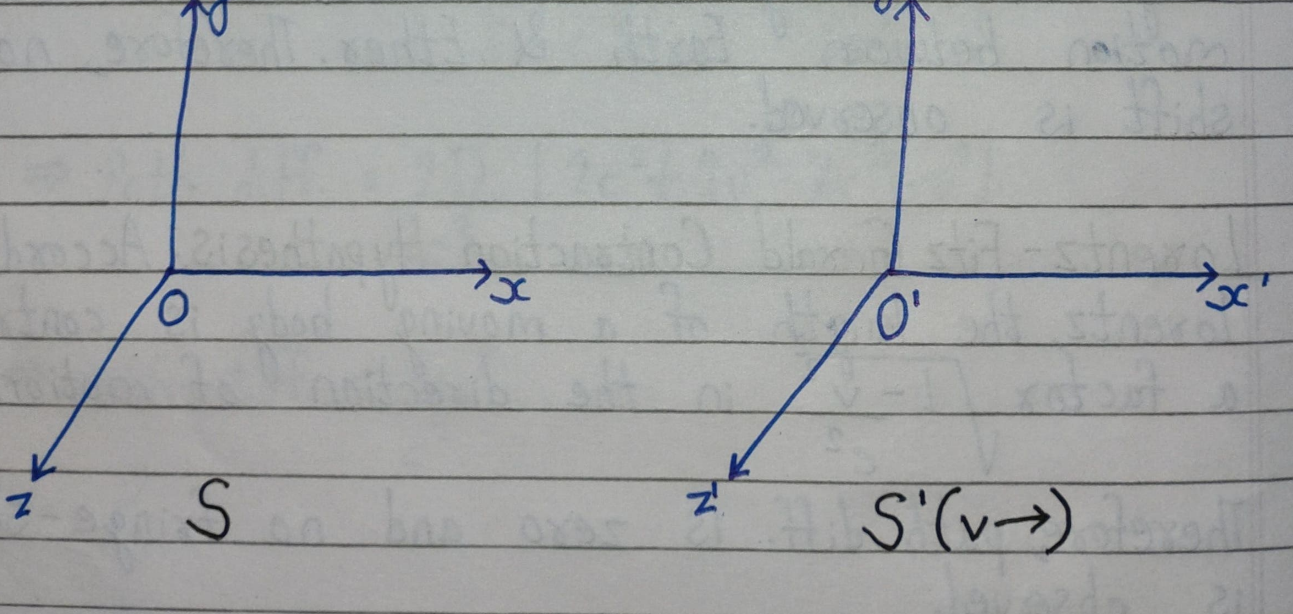
# Einstein's Postulates

1. First postulate states that, the laws of physics have same mathematical form in every inertial frame of reference.

2. Second postulate states that, speed of light is constant and does not depend upon motion of source, medium or observer.

# Lorentz Transformation Equations

Consider two inertial frames $S$ & $S'$. At $t=t'=0$, both frames work coincident. Frame $S$ is at rest and $S'$ moves with constant velocity $v$ in positive $x$-direction.



$$x' \propto (x - vt)$$

$$\Rightarrow x' = \gamma(x - vt) \text{ ——— } (1)$$

where,

$\gamma \rightarrow$ Constant of Proportionality.

Using First Postulate,

$$x = \gamma(x' + vt') \text{ ——— } (2)$$

Putting $x'$ in eq$^n$. (2),

$$x = \gamma[\gamma(x - vt) + vt']$$

$$\Rightarrow \frac{x}{\gamma} = \gamma x - \gamma vt + vt'$$

$$\Rightarrow vt' = \frac{x}{\gamma} - \gamma x + \gamma vt$$

$$\Rightarrow t' = \frac{x}{\gamma v} - \frac{\gamma x}{v} + \gamma t$$

$$\Rightarrow t' = \gamma t - \frac{\gamma x}{v} + \frac{x}{\gamma v}$$

$$\Rightarrow t' = \gamma t - \frac{\gamma x}{v}\left(1 - \frac{1}{\gamma^2}\right) \text{ ——— } (3)$$

Using First Postulate,

$$t = \gamma t' + \frac{\gamma x'}{v}\left(1 - \frac{1}{\gamma^2}\right) \text{ ——— } (4)$$